

Mikhail Razakov

Compiler & Program Analysis Engineer

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Compiler/program-analysis engineer with professional LLVM 22 and C#.NET static-analysis experience. The strongest public proof points are an interprocedural LLVM pass with explicit applicability boundaries, an end-to-end educational x86-64 backend, and Roslyn CFG fixed-point data flow.

EXPERIENCE

- 2026 — present** **ISP RAS — Static Analysis Engineer**
- Contribute to SharpChecker, ISP RAS's C#.NET static-analysis platform.
- July — August 2026** **MCST — Compiler Engineering Intern**
- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
 - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.
- 2023–2026** **Independent / freelance engineering — product systems**
- VpnMediator: subscription, payment, and access state, device limits, Linux deployment, recovery, and rollback.
 - Diploma.school: jQuery dropdown interactions and responsive navigation; Unity licensing: server-side activation limits and Unity client integration.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv, verify, structural expectations, idempotence, and observable before/after execution comparison.

x86-64 Compiler Backend & Register Allocation

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use

Microsoft.CodeAnalysis.CSharp.Testing.

SKILLS

Compiler / IR

C++23, LLVM 22/IR, APInt, interprocedural analysis, CFG/SSA, dominators, loops, LICM, legality

Program Analysis

C#.NET, Roslyn ControlFlowGraph, fixed-point data flow, conditional edges, loops/back edges, joins

Backend / Codegen

Rust, SysV x86-64, liveness/interference, register allocation, spills, phi lowering

Correctness / Tooling

LLVM Verifier, conservative rejection, regression checks, differential execution, Linux, CMake

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote